Comparing One-Stage and Two-Stage Architectures for Detecting Unknown Defects in Manufacturing

1. Introduction

Quality control (QC) is fundamental to manufacturing, ensuring both product reliability and operational safety. Because defective components incur high costs and potential regulatory risks, visual inspection is a critical step in the production line.

Traditionally, visual inspection has been performed by human operators. However, manual inspection is often time-consuming, prone to fatigue-induced errors, and difficult to scale to high-throughput production lines. To address these limitations, automated optical inspection (AOI) systems have been widely adopted to improve inspection speed, ensure consistent quality, and reduce reliance on manual labour.

Recent advances in deep learning have significantly improved the performance of AOI systems. Convolutional neural networks (CNNs) and object detection models, such as YOLO-based architectures, have demonstrated excellent accuracy in detecting and classifying surface defects under controlled conditions. When trained on sufficiently large and well-labelled datasets, these supervised models can learn highly discriminative decision boundaries for known defect types.

However, most deep-learning systems rely on a 'closed-set' assumption—the belief that every defect type encountered in production was already seen during training. In practice, factors like machine wear or process drift often introduce unforeseen defect types that the model is unprepared to handle. New defect types may emerge due to machine wear, changes in materials, process drift, or unforeseen failure modes. Under such conditions, closed-set models may either misclassify unknown defects as known categories or fail to detect them altogether.

This limitation motivates the study of **open-set defect detection**, where a system is required not only to classify known defect types accurately, but also to identify and flag previously unseen or unfamiliar defect patterns as unknown. Open-set recognition is more aligned with real-world manufacturing scenarios, particularly in safety-critical or high-reliability applications.

Several strategies have been proposed to address open-set conditions, including integrated open-set recognition mechanisms within classifiers and hybrid pipelines that combine anomaly detection with supervised classification. However, the **practical trade-offs** between these architectural strategies—particularly in terms of robustness, failure modes, and computational cost—have not been systematically documented in the manufacturing inspection context.

Project Overview and Objectives

This project investigates deep learning-based defect detection under open-set conditions and conducts a controlled comparison between two architectural paradigms:

1. **One-stage (integrated) open-set recognition**, where novelty detection is performed within a single classifier using feature-space analysis.
2. **Two-stage (cascaded) anomaly–classification pipelines**, where anomaly detection is used as a pre-filter before defect classification.

The objectives of this project are to:

- Evaluate the ability of each architecture to detect unseen defect types,
- Analyse system robustness and failure modes across different open-set configurations,
- Examine the computational and inference-time implications of each approach.

This interim report focuses on the **one-stage integrated open-set recognition architecture**, presenting methodology and experimental results on the NEU surface defect dataset.

2. Literature Review

2.1 Supervised Defect Detection in Manufacturing

Supervised learning has become the dominant paradigm for defect detection in industrial inspection tasks. Given sufficient labelled data, CNN-based models can learn highly discriminative representations that enable accurate classification of known defect categories. Object detection frameworks are particularly popular in manufacturing due to their ability to perform localisation and classification simultaneously, while satisfying real-time constraints.

Lei et al. [1] proposed a YOLO-based defect detection system that integrates feature extraction and classification for industrial surface inspection. Their results demonstrate strong performance on known defect classes, highlighting the effectiveness of supervised deep learning approaches in controlled settings. Similar works report near-perfect accuracy when training and test data are drawn from the same set of defect categories.

However, supervised defect detectors inherently assume that all defect types encountered during deployment are present in the training data. When this assumption is violated, model predictions can become unreliable. As such, while supervised detectors provide a strong baseline, they are fundamentally limited in open-set environments.

2.2 Anomaly Detection for Visual Inspection

Anomaly detection represents an alternative paradigm for industrial inspection, focusing on modelling normality rather than explicitly classifying defect types. These methods are typically trained using only defect-free samples and aim to identify deviations from learned normal patterns.

PatchCore, introduced by Roth et al. [2], is a state-of-the-art anomaly detection method that operates by extracting patch-level features from convolutional backbones and performing nearest-neighbour matching against a memory bank of normal features. PatchCore has demonstrated strong performance on industrial inspection benchmarks, including MVTec AD, particularly for texture-based defects.

Anomaly detection methods offer several advantages: they do not require labelled defect data, are naturally suited to detecting previously unseen defects, and align well with scenarios where defect diversity is large or unpredictable. However, they lack semantic classification capabilities and typically cannot distinguish between different types of defects.

Zavrtanik et al. [3] proposed DRAEM, which augments anomaly detection through reconstruction-based learning. While effective, such methods remain limited to binary normal-versus-abnormal decisions.

2.3 Open-Set and Hybrid Defect Detection Approaches

To bridge the gap between supervised defect detection and anomaly detection, hybrid approaches have been proposed to handle open-set conditions.

2.3.1 Integrated Open-Set Recognition in Defect Detection

Integrated open-set recognition extends supervised classifiers by incorporating novelty detection mechanisms, such as confidence estimation or feature-space distance measurements. In this approach, feature embeddings extracted from a trained classifier are compared against stored representations of known classes. Samples that lie sufficiently far from known distributions are flagged as unknown.

Lei et al. [1] also explored feature-based rejection mechanisms, illustrating how distance measures can be used to identify unfamiliar patterns. This category of methods is referred to in this project as a **one-stage architecture**, as classification and novelty detection are performed within a single model.

While computationally efficient, integrated OSR methods are sensitive to feature overlap between known and unknown classes and may fail when unknown defects closely resemble known ones.

2.3.2 Cascaded Anomaly–Classification Pipelines

An alternative strategy employs anomaly detection as a pre-filter, followed by supervised classification. Cheng and Yeh [4] proposed a cascaded pipeline in which an anomaly detector identifies abnormal samples, which are then passed to a classifier for semantic labelling. This design prioritises robustness and safety by ensuring that unknown or abnormal samples are detected before classification.

Such **two-stage architectures** are more computationally expensive and require additional data for anomaly detection training, but are potentially more reliable under open-set conditions.

2.4 Summary and Research Gap

Existing literature demonstrates that supervised defect detectors achieve high accuracy but assume closed-set conditions. Anomaly detectors handle novelty well but cannot provide semantic labels. Hybrid approaches exist but are often evaluated in isolation.

There is limited work that systematically compares integrated open-set recognition and cascaded anomaly–classification pipelines under controlled conditions, particularly with respect to robustness, failure modes, and computational cost in manufacturing inspection scenarios.

2.5 Final Takeaway

There remains a lack of controlled, system-level evaluation comparing different architectural strategies for open-set defect detection in manufacturing environments. This project aims to address this gap.

3. Methodology

We describe the experimental methodology used to evaluate the integrated open-set recognition (OSR) architecture for defect detection under open-set conditions. The focus is on ensuring a controlled, reproducible evaluation that isolates the behaviour of the one-stage OSR approach before introducing more complex architectures in later stages of the project.

3.1 Dataset Description

Experiments use the **NEU Surface Defect Dataset**, which contains images from six defect categories: Crazing, Inclusion, Patches, Pitted Surface, Rolled-in Scale, and Scratches. Although the NEU dataset includes predefined folders, we pooled all available images to

implement a standardized open-set protocol. This ensures that our experimental splits remain consistent and reproducible, regardless of the original data structure.

For each open-set split, images belonging to known defect classes are re-partitioned using a **stratified 70/15/15 split** into training, validation (calibration), and test sets, based on a fixed random seed. Images belonging to unknown defect classes are excluded from training and calibration and are used **only for testing**.

All splits therefore follow an identical protocol regardless of the dataset's original folder structure, avoiding bias from predefined partitions not designed for open-set evaluation.

The NEU dataset contains only defective images and does not include defect-free ("good") samples. Consequently, the evaluation focuses on **open-set classification among defect types**, rather than defect-versus-normal detection.

3.2 Open-Set Experimental Protocol

Three predefined class splits (A/B/C) define known vs unknown defect categories.

Each split assigns a disjoint subset of defect classes as *known* during training, while the remaining classes are treated as *unknown* and appear only at test time.

The splits are defined as follows.

1. **Split A**
   Known classes: Crazing, Inclusion, Patches
   Unknown classes: Pitted Surface, Rolled-in Scale, Scratches
2. **Split B**
   Known classes: Scratches, Rolled-in Scale, Pitted Surface
   Unknown classes: Crazing, Inclusion, Patches
3. **Split C**
   Known classes: Crazing, Scratches, Patches
   Unknown classes: Inclusion, Pitted Surface, Rolled-in Scale

Using multiple predefined splits allows the evaluation to capture different visual relationships between known and unknown defect types, and prevents conclusions from being drawn based on a single favourable or unfavourable configuration.

For each split:

1. Known classes are used for training, calibration, and testing.
2. Unknown classes are excluded from training and calibration and appear only at test time.

3.2.1 Training, Calibration, and Test Sets

For each split:

1. Known-class train/val/test: all known-class images (from both NEU train and validation folders) are stratified into 70% train, 15% validation (calibration), and 15% test, using a fixed random seed to preserve class balance.
2. Unknown-class test: all images from the unknown classes (across both NEU folders) are used as test-only samples.

This procedure ensures strict separation: unknown classes are never used in training or threshold calibration.

3.3 One-Stage Integrated Open-Set Recognition Architecture

The integrated OSR architecture extends a standard supervised classifier with a feature-space novelty detection mechanism. Classification and unknown detection are performed within a single model, without the use of external anomaly detectors.

3.4 Classifier Architecture and Training

A ResNet-50 convolutional neural network is used as the base classifier due to its strong representational capacity and widespread adoption in visual inspection tasks. The network is initialised with ImageNet-pretrained weights, and the final fully connected layer is replaced to match the number of known defect classes for each split.

The classifier is trained **only on known classes** using the cross-entropy loss function. Optimisation is performed using the Adam optimiser. **No data augmentation is applied**, aside from resizing and normalisation, in order to preserve the intrinsic visual characteristics of defect patterns.

We train for 15 epochs by default. The **best model checkpoint is selected based on validation accuracy** and is used for all subsequent embedding extraction and open-set evaluation.

3.5 Feature Embedding Extraction

To enable open-set recognition, feature embeddings are extracted from the **penultimate layer** of the trained classifier. Specifically, embeddings are taken after the global average pooling layer and before the final classification layer, resulting in a fixed-dimensional representation for each image.

These embeddings capture high-level semantic information in a compact form suitable for distance-based analysis, and are commonly used in open-set detection literature.

We extract features with the network in evaluation mode, without data augmentation, ensuring deterministic and consistent representations.

3.6 Open-Set Recognition via Mahalanobis Distance

For each known defect class, a **class-conditional Gaussian distribution** is fitted to the corresponding training embeddings. This involves estimating:

- the mean feature vector, and
- the covariance matrix of the embeddings.

To ensure numerical stability, covariance matrices are regularised by adding a small multiple of the identity matrix.

At inference time, the **Mahalanobis distance** between a test embedding and each known class distribution is computed. The minimum distance across all known classes is used as the open-set score:

- Low distance indicates similarity to a known defect class.
- High distance indicates deviation from known feature distributions.

Novelty detection is thus performed without altering the model architecture or adding training parameters.

3.7 Confidence-Based Rejection

In addition to distance-based rejection, a **confidence gate** based on the maximum softmax probability is applied. Softmax confidence provides a complementary signal that reflects the classifier's certainty in its prediction.

A test sample is flagged as unknown if:

- its Mahalanobis distance exceeds a calibrated threshold, **or**
- its classification confidence falls below a calibrated confidence threshold.

This combined criterion aims to capture both geometric deviation in feature space and uncertainty in classifier predictions.

3.8 Threshold Calibration Strategy

All thresholds are calibrated using **known-class validation data only**, in order to avoid test-set leakage.

For both distance-based and confidence-based rejection:

- thresholds are selected such that **95% of known validation samples are accepted** as known.

This operating point reflects a conservative industrial setting where false rejections of known defect types are minimised, while still enabling unknown detection.

Threshold-independent performance is additionally assessed using the area under the receiver operating characteristic curve (AUROC).

3.9 Evaluation Metrics

We evaluate performance using the following metrics:

- **Known-class accuracy and macro-F1 score**, measured on known-class test samples only.
- **AUROC**, measuring separability between known and unknown samples.
- **True positive rate for unknown detection (TPR_unknown)** at the calibrated operating point.
- **False positive rate on known samples (FPR_known)** at the calibrated operating point.

These metrics jointly capture classification accuracy, novelty detection capability, and robustness under open-set conditions.

3.10 Summary of Methodology

The proposed methodology enables a controlled evaluation of integrated open-set recognition by isolating novelty detection within a single classifier and systematically varying known and unknown class configurations. By enforcing strict data separation and validation-based threshold calibration, the experimental setup ensures that observed performance reflects genuine generalisation rather than artefacts of data leakage or overfitting.

This methodological foundation provides a clear baseline against which more complex, cascaded architectures can be compared in subsequent stages of the project.

4. Experimental Results and Analysis

We present experimental results for the integrated open-set recognition (OSR) architecture on the NEU Surface Defect Dataset. Results are reported across three predefined open-set configurations (Split A, Split B, and Split C) to evaluate robustness under varying known–unknown class relationships. All experiments follow the methodology described in Section 3.

For each split, results are presented in two groups of figures:

1. **Training behaviour**, showing classifier convergence (accuracy and loss).
2. **Open-set recognition behaviour**, showing novelty separability (Mahalanobis distance histogram and ROC curve).

4.1 Results for Split A

Split A represents a challenging open-set configuration in which unknown defect classes share substantial visual similarity with the known classes.

4.1.1 Classification Performance on Known Classes

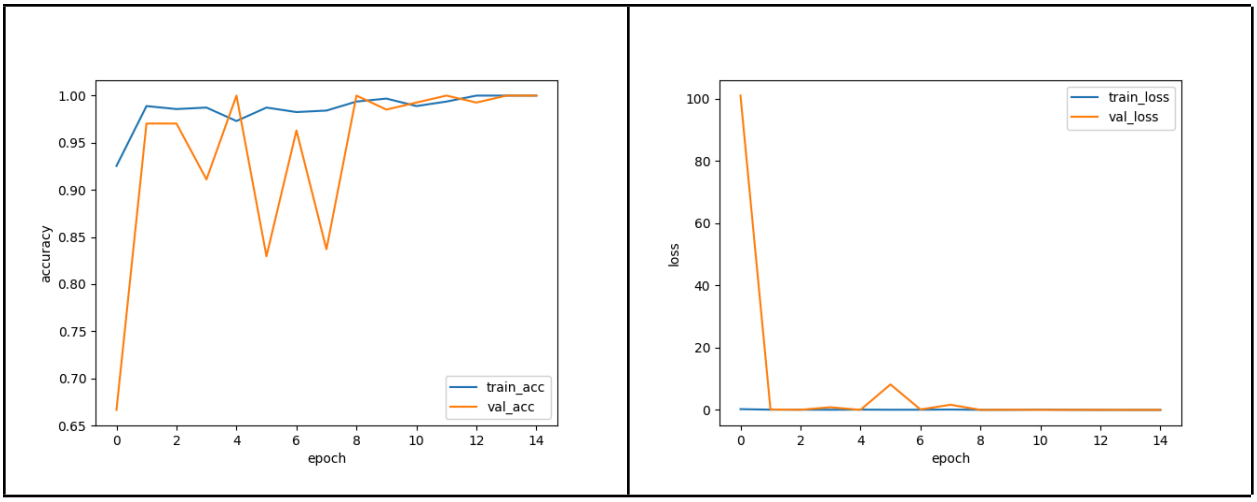

Figure 4.1: Training dynamics for Split A. (a) Training and validation accuracy. (b) Training and validation loss.

The classifier converges rapidly, achieving near-perfect accuracy on both training and validation sets. Training and validation loss decrease smoothly and remain stable after convergence. The classifier learns highly discriminative representations for known defect classes and generalises well under closed-set conditions.

The absence of overfitting or instability suggests that degraded open-set performance stems from architectural limitations rather than poor classifier training.

4.1.2 Open-Set Recognition Performance

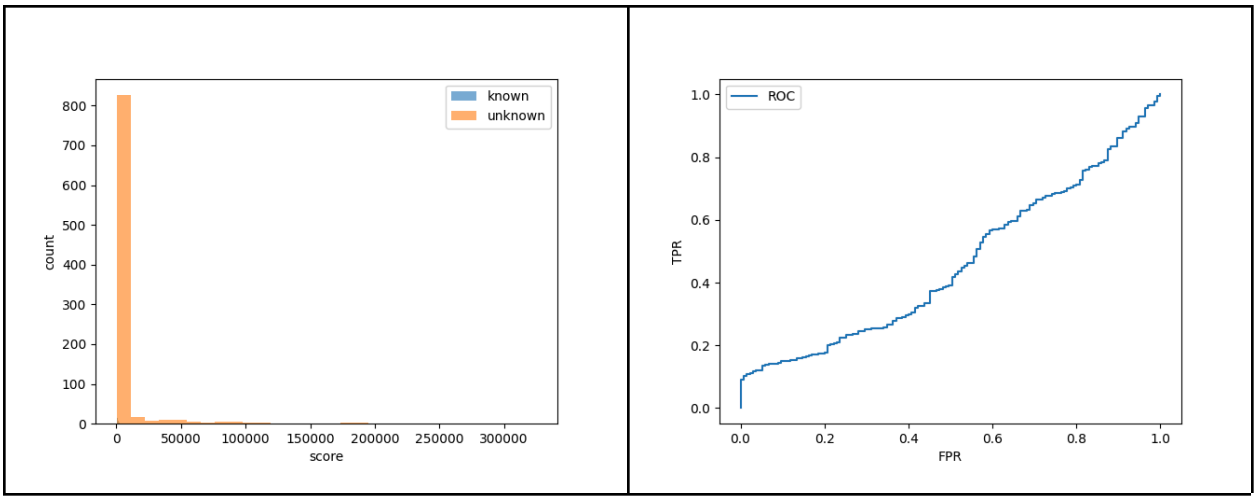

Figure 4.2: Open-set recognition performance for Split A. (a) Mahalanobis distance distributions for known and unknown samples. (b) ROC curve for known vs unknown detection.

Despite strong closed-set performance, the OSR mechanism performs poorly under Split A. The Mahalanobis distance distributions for known and unknown samples exhibit substantial overlap, indicating that many unknown samples lie close to known feature distributions. The ROC curve yields an AUROC of approximately 0.46, close to random chance.

At the calibrated operating point, a relatively high true positive rate for unknown detection is achieved, but this comes at the cost of an elevated false positive rate on known samples. Threshold-based rejection proves fragile in this configuration.

4.1.3 Discussion

Split A reveals a significant failure mode: when unknown defects visually resemble known classes, they occupy the same regions in the model's feature space.

Distance-based metrics cannot separate them when the classifier performs near-perfectly on known categories.

4.2 Results for Split B

Split B represents a favourable open-set configuration in which unknown defect classes are visually distinct from the known classes.

4.2.1 Classification Performance on Known Classes

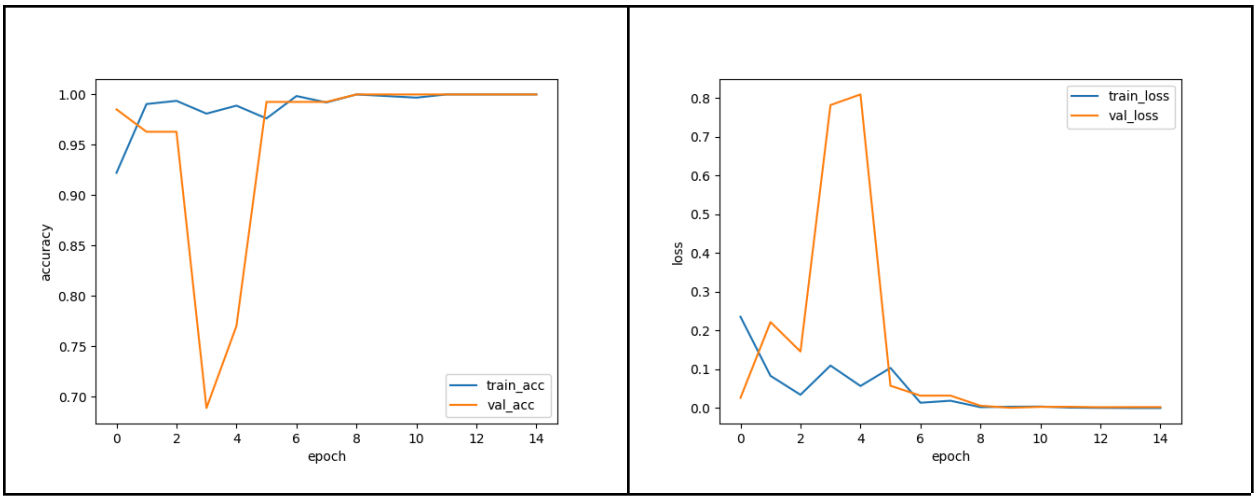

Figure 4.3: Training dynamics for Split B. (a) Training and validation accuracy. (b) Training and validation loss.

Training and validation accuracy converge to near-perfect values after minor early-epoch fluctuations. Loss curves decrease steadily and remain stable, confirming consistent optimisation and good generalisation on known classes.

4.2.2 Open-Set Recognition Performance

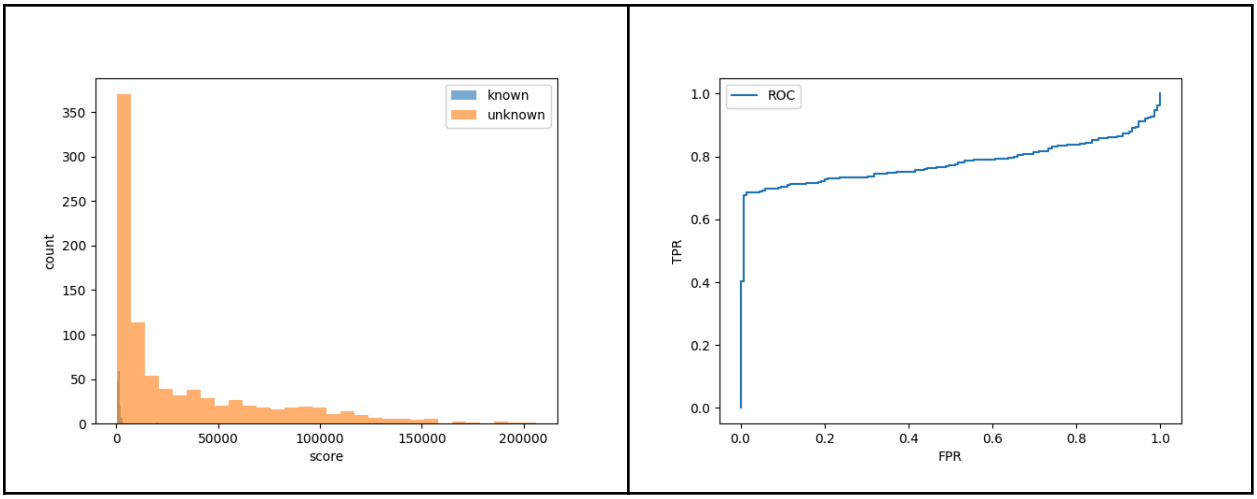

Figure 4.4: Open-set recognition performance for Split B. (a) Mahalanobis distance distributions for known and unknown samples. (b) ROC curve for known vs unknown detection.

In contrast to Split A, unknown samples in Split B tend to produce significantly higher Mahalanobis distances than known samples, with noticeably reduced overlap between distributions. The ROC curve exhibits a steep rise near the origin, yielding an AUROC of approximately 0.78.

Integrated OSR reliably distinguishes unknown defects when unknown classes differ substantially from known classes in visual appearance.

4.2.3 Discussion

Split B demonstrates that integrated open-set recognition can perform effectively under favourable conditions. However, this success depends strongly on the visual separability of known and unknown defect types, rather than on inherent robustness of the architecture.

4.3 Results for Split C

Split C is a mixed open-set configuration designed to approximate a more realistic deployment scenario.

4.3.1 Classification Performance on Known Classes

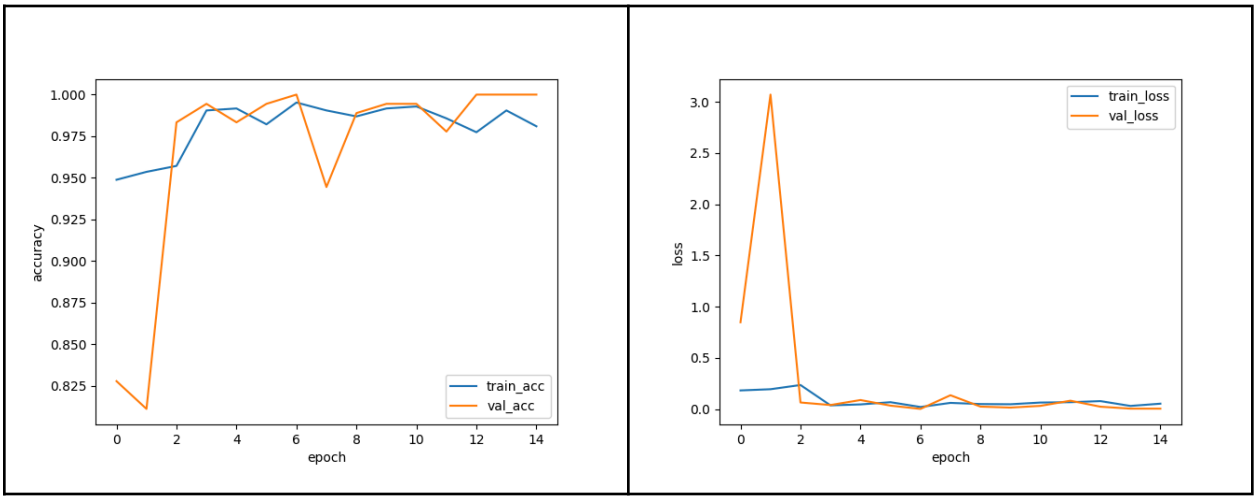

Figure 4.5: Training dynamics for Split C. (a) Training and validation accuracy. (b) Training and validation loss.

The classifier maintains strong performance, with training accuracy stabilising near 98–99% and validation accuracy converging to approximately 100%. Training and validation loss curves remain well aligned after convergence, indicating stable optimisation.

4.3.2 Open-Set Recognition Performance

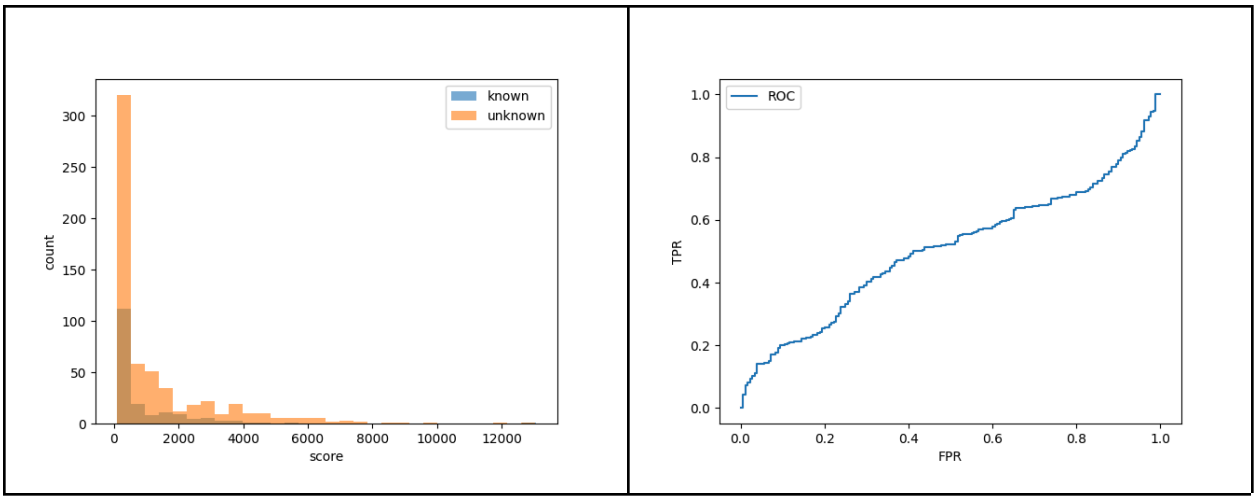

Figure 4.6: Open-set recognition performance for Split C. (a) Mahalanobis distance distributions for known and unknown samples. (b) ROC curve for known vs unknown detection.

While unknown samples generally produce higher distance scores than known samples, substantial overlap remains between the two distributions. The ROC curve yields an AUROC close to 0.5, indicating weak separability and high sensitivity to threshold selection.

4.3.3 Discussion

Split C highlights the limitations of integrated OSR in mixed scenarios where unknown defects partially overlap with known feature distributions. Although some novelty signal is present, separation is insufficiently robust for reliable deployment.

4.4 Cross-Split Comparison

Across all three splits, integrated open-set recognition exhibits substantial variability in performance. Split B represents a best-case scenario, Split A exposes a severe failure mode, and Split C demonstrates fragile average-case behaviour. These variations occur despite consistently high closed-set classification accuracy, indicating that classification success alone does not guarantee reliable unknown detection.

4.5 Quantitative Analysis

Split	AUROC	$TPR_{unknown}$	FPR_{known}	Open-set Accuracy	Known Accuracy
Split A	0.460	0.651	0.082	0.686	1.000
Split B	0.779	0.972	0.022	0.973	0.993

Split C	0.506	0.960	0.133	0.939	1.000
Mean	**0.582**	**0.861**	**0.079**	**0.866**	**0.998**
Std	**0.141**	**0.149**	**0.045**	**0.128**	**0.004**

Table 4.1: Open-set recognition performance of the integrated OSR architecture on the NEU Surface Defect Dataset across three predefined splits.
Metrics are reported at a calibrated operating point corresponding to 95% acceptance of known validation samples. Mean and standard deviation are computed across splits.

As shown in Table 4.1, AUROC ranges from near-random performance in Split A (0.46) to substantially stronger separability in Split B (0.78). This variation occurs despite minimal differences in known-class accuracy across splits, indicating that classification performance alone is not a reliable indicator of open-set robustness.

Although high true positive rates for unknown detection are achieved in some configurations, these gains are often accompanied by elevated false positive rates on known samples, particularly in Split C. This trade-off exposes the sensitivity of threshold-based rejection and the fragility of one-stage open-set recognition under challenging conditions.

Overall, the results demonstrate that integrated OSR can perform effectively under favourable circumstances but lacks consistent robustness across diverse open-set scenarios. This variability motivates the exploration of alternative architectural strategies that prioritise reliable unknown detection.

5. Discussion and Interim Conclusion

The quantitative results in Table 4.1 reinforce the qualitative observations presented in Section 4. Despite near-perfect classification accuracy on known defect classes across all splits, open-set detection performance exhibits high variability. This discrepancy highlights a fundamental limitation of one-stage open-set recognition approaches: strong closed-set classification does not guarantee reliable detection of unknown defects.

The results demonstrate that one-stage OSR approaches can perform well when unknown defects are visually distinct, but fail when unknown defects resemble known classes in feature space. In such cases, distance-based novelty detection becomes unreliable despite high classification confidence.

From a practical perspective, this sensitivity raises concerns for manufacturing inspection systems deployed in safety-critical environments, where the nature of unknown defects cannot be assumed in advance. The findings suggest that integrated OSR alone may be insufficient to guarantee robust unknown defect detection.

Motivated by these observations, the next phase of this project will investigate a **two-stage cascaded anomaly–classification architecture**, in which anomaly detection explicitly models abnormality prior to semantic classification. The final study will compare the robustness and computational cost of the two architectures to assess whether the additional complexity of a cascaded approach is justified.

References

[1] X. Lei, J. Zhang, Y. Liu, and H. Wang, “Surface defect detection based on YOLO and feature extraction,” *Sensors*, vol. 24, no. 3, pp. 1–18, 2024.

[2] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, “Towards total recall in industrial anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 14318–14328.

[3] V. Zavrtanik, M. Kristan, and D. Skočaj, “Reconstruction-based anomaly detection for surface inspection,” *Pattern Recognition*, vol. 120, 2021.

[4] C. Cheng and C. Yeh, “Hybrid anomaly detection and classification for industrial inspection,” *IEEE Access*, 2023.